

A Practical Guide to BLEU score

TOPIC -- BLEU SCORE

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Brevity penalty The modified n-gram precision unduly gives a high score for candidate strings that are "telegraphic", that is, containing all the n-grams of the reference strings, but for as few times as possible. In order to punish candidate strings that are too short, define the brevity penalty to be $BP(\hat{S}; S) := e^{-(r/c - 1)^+}$ where $(r/c - 1)^+ = \max(0, r/c - 1)$ is the positive part of $r/c - 1$.

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r is the effective reference corpus length, that is, $r := \sum_{i=1}^M |y^{(i,j)}|$ where $y^{(i,j)} = \arg \min_{y \in S_i} |y| - |y^{(i)}|$, that is, the sentence from S_i whose length is as close to $|y^{(i)}|$ as possible.

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c is the length of the candidate corpus, that is, $c := \sum_{i=1}^M |\hat{y}^{(i)}|$ where $|\hat{y}|$ is the length of y .

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where m is number of words from the candidate that are found in the reference, and w_t is the total number of words in the candidate. This is a perfect score, despite the fact that the candidate translation above retains little of the content of either of the references. The modification that BLEU makes is fairly straightforward. For each word in the candidate translation, the algorithm takes its maximum total count, $m_{\max}$, in any of the reference translations. In the example above, the word "the" appears twice in reference 1, and once in reference 2. Thus $m_{\max} = 2$. For the candidate translation, the count m_w of each word is clipped to a maximum of $m_{\max}$ for that word. In this case, "the" has $m_w = 7$ and $m_{\max} = 2$, thus m_w is clipped to 2. These clipped counts m_w are then summed over all distinct words in the candidate. This sum is

then divided by the total number of unigrams in the candidate translation. In the above example, the modified unigram precision score would be:

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This quantity measures how many n-grams in the reference sentence are reproduced by the candidate sentence. Note that we count the n-substrings, not n-grams. For example, when $\hat{y} = a b a$, $y = a b a b a b a$, $n = 2$ $\{\hat{y}\} = aba, y = abababa, n = 2$, all the 2-substrings in $\hat{y}$ (ab and ba) appear in y 3 times each, so the count is 6, not 2. In the above situation, however, the candidate string is too short. Instead of 3 appearances of ab it contains only one, so we add a minimum function to correct for that: $\sum_{s \in G_n(\hat{y})} \min(C(s, \hat{y}), C(s, y))$ $\{\sum_{s \in G_n(\hat{y})} \min(C(s, \hat{y}), C(s, y))\}$ This count summation cannot be used to compare between sentences, since it is not normalized. If both the reference and the candidate sentences are long, the count could be big, even if the candidate is of very poor quality. So we normalize it $\sum_{s \in G_n(\hat{y})} \min(C(s, \hat{y}), C(s, y)) / \sum_{s \in G_n(\hat{y})} C(s, \hat{y})$ $\{\frac{\sum_{s \in G_n(\hat{y})} \min(C(s, \hat{y}), C(s, y))}{\sum_{s \in G_n(\hat{y})} C(s, \hat{y})}\}$ The normalization is such that it is always a number in $[0, 1]$, allowing meaningful comparisons between corpuses. It is zero if none of the n-substrings in candidate is in reference. It is one if every n-gram in the candidate appears in reference, for at least as many times as in candidate. In particular, if the candidate is a substring of the reference, then it is one.

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When $r \leq c$, the brevity penalty $BP = 1$, meaning that we don't punish long candidates, and only punish short candidates. When $r > c$, the brevity penalty $BP = e^{1 - r/c}$

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BLEU (bilingual evaluation understudy) is an algorithm for evaluating the quality of text which has been machine-translated from one natural language to another. Quality is considered to be the correspondence between a machine's output and that of a human: "the closer a machine translation is to a professional human translation, the better it is" – this is the central idea behind BLEU.[1] Invented at IBM in 2001, BLEU was one of the first metrics to claim a high correlation with human judgements of quality,[2][3] and remains one of the most popular automated and inexpensive metrics. Scores are calculated for individual

translated segments—generally sentences—by comparing them with a set of good quality reference translations. Those scores are then averaged over the whole corpus to reach an estimate of the translation's overall quality. Intelligibility or grammatical correctness are not taken into account.[4] BLEU's output is always a number between 0 and 1. This value indicates how similar the candidate text is to the reference texts, with values closer to 1 representing more similar texts. Few human translations will attain a score of 1, since this would indicate that the candidate is identical to one of the reference translations. For this reason, it is not necessary to attain a score of 1. Because there are more opportunities to match, adding additional reference translations will increase the BLEU score.[5]